Quick Guide to HMW90 Maintenance & Calibration

What This Device Does

The HMW90 is a transmitter that measures humidity (RH) and temperature (T). Over time, these measurements may drift and need adjustment to stay accurate.

Three Ways to Adjust Your Device

1. Using the Built-in Trimmers (Simplest Method)

- Open the sliding cover on the device
- Turn the RH or T trimmer knob gently (it only rotates 135° - don't force it!)
- The screen shows you're adjusting - turn the knob to set your correction
- Stop turning and wait for it to save
- You can repeat this process if you need more correction - adjustments add up

2. Using the HM70 Handheld Meter

- Connect your HMW90 to the HM70 using Vai's connection cable
- Go to Functions menu → Calibrate → Start
- Choose RH or T to adjust
- Pick your adjustment style:
  - Match a reference: Makes your device read the same as a connected reference sensor
  - 1-point adjustment: You tell it what the correct value should be
  - 2-point adjustment: Adjust at two different humidity levels (RH only)
  - Reset to factory: Removes your custom adjustments
- Follow the on-screen instructions

3. Using a Computer (Most Detailed)

- Connect with Vai USB cable to your computer

- Open terminal software (settings: 19200, N, 8, 1)

- Check current settings with commands: `pass 9000`, `crh`, `ct`

- Place the device in a known reference environment and let it stabilize

- Use `CRH` and `CT` commands to make adjustments

- Verify the readings are now correct

Important Things to Know

- Humidity adjustments are smart: A 1-point RH adjustment automatically corrects both offset and gain based on the humidity level you're adjusting at

- Temperature adjustments are simple: Just a basic offset correction

- Your adjustments are saved: They're stored in the measurement module, so replacing the module means starting fresh

- Minimum requirement: For RH adjustment, the target value must be at least 50% of what's currently displayed (prevents extreme corrections)

When adjustment isn't enough, you can replace the entire measurement module to restore accuracy.